CONFOEDERATIO,
RESEARCH DIVISION (CRD).

DEMOGRAPHY AND SETTLEMENT OF THE POST-SOVIET SPACE.

A Regional Literature Review of Population and Settlement (1991AD - 2022AD).

10 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

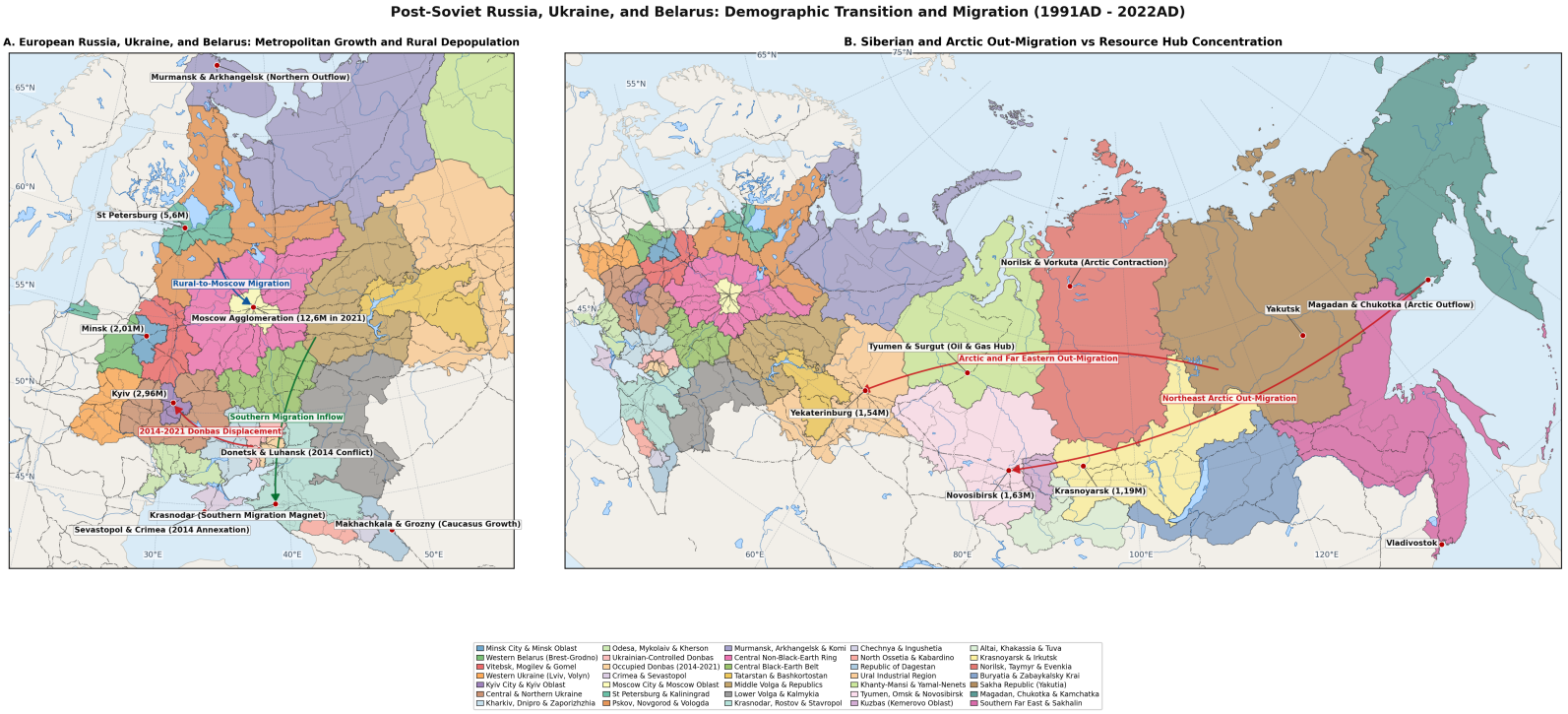


Figure 1. Thirty-five post-Soviet demographic regions showing metropolitan growth poles, northern shrinkage, armed conflicts and migration flows (1991AD - 2021AD).

Background

The dissolution of the Union of Soviet Socialist Republics in December 1991AD inaugurated a thirty-year demographic transformation across the East Slavic and Northern Eurasian space comprising the Republic of Belarus, Ukraine and the Russian Federation (1991-2021AD). At the 1991AD transition from the late Soviet era (224,55M across the seven European and Northern Eurasian Union Republics), the restoration of independence to the Estonian (1,57M), Latvian (2,66M), Lithuanian (3,7M) and Moldavian (4,36M) Republics removed 12,29M outside the post-Soviet RU-UA-BY study envelope, leaving a 1991AD baseline population of 212,26M (212,26M) across the 35 demographic regions (17.969.007 km2: 10,25M across Regions 1-3 in Belarus, 51,5M across Regions 4-11 within the internationally recognised 1991AD borders of Ukraine and 150,51M across Regions 12-35 in the Russian Federation). This total peaked briefly at 212,82M (212,82M) in 1995AD during the initial wave of post-Soviet repatriation from Central Asia and the South Caucasus [1][2][3][4]. Over the ensuing quarter-century (1995-2021AD), this territorial expanse experienced a steep peacetime demographic contraction alongside a marked spatial reordering of its

urban and rural settlement hierarchy. By late 2021AD - on the eve of the full-scale Russian invasion of Ukraine in February 2022AD - the combined population across the 35 regions had fallen by -14,18M (-14,18M, or -6,7% relative to 1991AD) to 198,08M (198,08M), comprising 9,15M (-10,7%) in Belarus (Regions 1-3), 41,56M (-19,3%) across the 1991AD territories of Ukraine (Regions 4-11, including 2,21M in Government-Controlled Donbas Reg. 9, 2,47M in Pre-2022 Occupied Donbas Reg. 10 and 2,44M in Annexed Crimea and Sevastopol Reg. 11) and 147,37M (-2,1%) across the Russian Federation (Regions 12-35) [3][4][5][6].

Methodology

Population series for 1991AD - 2021AD rest on three national statistical systems that diverged in census frequency and territorial coverage after 1991AD [7].

In the Russian Federation (Regions 12-35, Goskomstat Rossii/Rosstat), following the 1989AD All-Union Census (147,02M permanent residents, rising to 150,51M in the 1991AD regional baseline and 151,39M in 1995AD), Rosstat conducted three post-Soviet national enumerations: the All-Russian Population Census of 9 October 2002AD (145,17M) [8], the All-Russian Population Census of 14 October 2010AD (142,86M) [9] and the delayed All-Russian Population Census of 1 October - 14 November 2021AD (147,18M including Crimea/Sevastopol, or 147,37M within Regions 12-35) [5][10]. As demographic analysts led by Anatoly Vishnevsky, Sergei Zakharov, Evgueni Andreev and Nikita Mkrtchyan have demonstrated, the 2002AD and 2021AD censuses relied heavily on administrative police-registration records (*registratsiya*) in metropolitan Moscow (Region 12), Saint Petersburg (Region 13) and the North Caucasus (Regions 22-24), requiring careful cross-checking against civil vital-registration ledgers (ZAGS births and deaths), intercensal retrospective adjustments and municipal migration matrices [11][12][13][14].

In Ukraine (Regions 4-11, Derzhstat Ukrainy and the Ptoukha Institute), after the 1989AD Soviet census (51,45M permanent residents, rising through CIS return migration to 52,2M in 1993AD), independent Ukraine conducted only a single traditional door-to-door enumeration, the First All-Ukrainian Population Census of 5 December 2001AD (48,46M) [15]. Following the March 2014AD Russian annexation of Crimea and Sevastopol (Region 11) and the outbreak of the War in Donbas (April 2014AD), which split Donetsk and Luhansk oblasts along a 427 km Line of Contact between Government-Controlled Donbas (Region 9) and Pre-2022 Occupied Donbas (Region 10), Ukrainian demographic series were maintained through annual Ukrstat vital-registration balances [3], Ptoukha Institute reconstructions (Libanova, Gladun, Levchuk) and the December 2019AD Electronic Population Assessment (37,29M on government-controlled territory, or 41,56M across all eight Ukrainian regions, Regions 4-11, in 2021AD) [6][16][17][18].

In the Republic of Belarus (Regions 1-3, Minstat/Belstat), national statistical authorities conducted full decennial enumerations on 16 February 1999AD (10,04M) [19], 14 October 2009AD (9,5M) [20] and 4-30 October 2019AD (9,41M, declining to 9,15M present population across Regions 1-3 by late 2021AD under COVID-19 excess mortality and post-2020AD emigration) [4][21][22].

Rather than grouping the post-Soviet space into broad pre-industrial ecological zones, this study divides Belarus (Regions 1-3), Ukraine (Regions 4-11) and the Russian Federation (Regions 12-35) into 35 demographic regions. In the North Caucasus, this division distinguishes three ethno-demographic regimes: Chechnya and Ingushetia (Region 22, shaped by the First and Second Chechen Wars of 1994-2009AD, the Nazran IDP surge and the subsequent post-war rebound), North Ossetia (site of the 1992AD Prigorodny conflict), Kabardino-Balkaria and Karachay-Cherkessia (Region 23) and the Republic of Dagestan (Region 24, marked by high-fertility natural increase and the Makhachkala-Kaspiysk urban boom). In Ukraine, the regional framework separates the three 2014-2021AD conflict and annexation zones: Government-Controlled Donbas (Region 9), Pre-2022 Occupied Donbas (Region 10) and Annexed Crimea and Sevastopol (Region 11). Across Asian Russia, ten regions (Regions 26-35) distinguish the

West-East Trans-Siberian and Pacific corridors (the West Siberian Trans-Siberian Belt [Reg. 27], the Kuzbass Coal Basin [Reg. 28], the East Siberian Trans-Siberian Corridor [Reg. 30], Buryatia and Zabaykalsky [Reg. 32] and the Southern Far East and Pacific Corridor [Reg. 35]) from the Southern Siberian Indigenous Republics (Altai, Khakassia and Tuva [Reg. 29]) and the Northern Hydrocarbon and Polar Zones (the Yugra and Yamal Hydrocarbon Arctic [Reg. 26], Norilsk, Taymyr and Evenkia [Reg. 31], the Sakha Republic/Yakutia [Reg. 33] and the Extreme Northeast Arctic Exodus of Magadan, Chukotka and Kamchatka [Reg. 34]).

Results

Regional Population Trajectories (1991-2021AD)

Figure 2 illustrates the trajectories of ten representative regions alongside the indexed national curves (1991AD = 100), while Table 1 lists the benchmark series (1991AD, 1995AD, 2000AD, 2005AD, 2010AD, 2013AD, 2015AD and 2021AD) across all 35 regions.

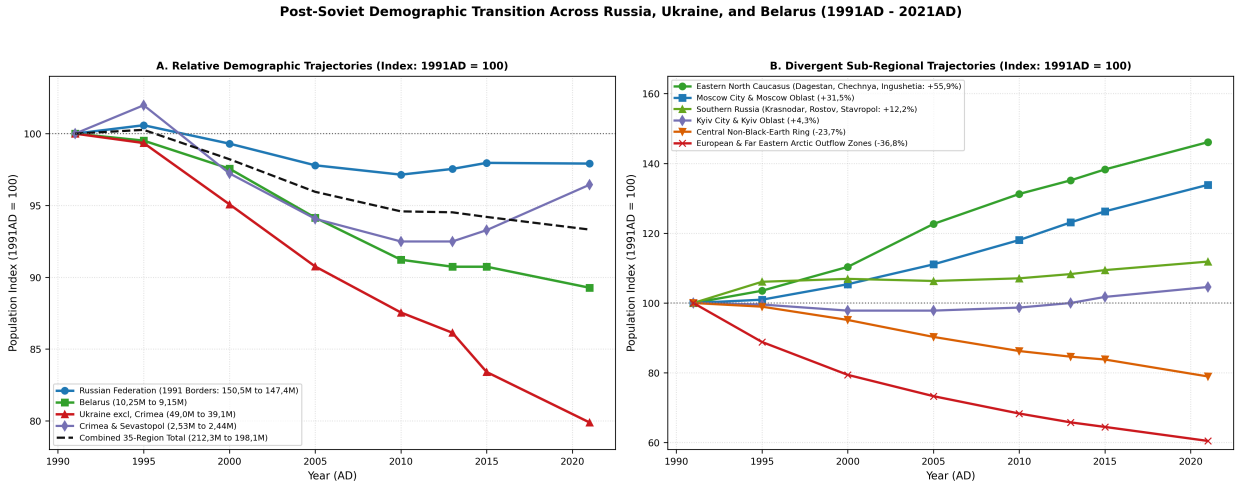


Figure 2. Regional population trajectories of expanding metropolises and contracting industrial and conflict zones alongside national indices (1991AD - 2021AD).

Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
1. Minsk City & Minsk Oblast Metropolitan Magnet	3,4M	3,39M	3,36M	3,32M	3,34M	3,38M	3,42M	3,48M	+2,4%
2. Western Belarus: Brest & Grodno Oblasts	2,68M	2,69M	2,65M	2,56M	2,47M	2,44M	2,43M	2,36M	-11,9%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
3. Eastern & Southeastern Belarus: Vitebsk, Mogilev & Gomel Oblasts	4,17M	4,12M	3,99M	3,77M	3,54M	3,48M	3,45M	3,31M	-20,6%
4. Western Ukraine	10,25M	10,28M	10,05M	9,78M	9,62M	9,58M	9,56M	9,32M	-9,1%
5. Kyiv City & Kyiv Metropolitan Oblast	4,58M	4,56M	4,48M	4,48M	4,52M	4,58M	4,66M	4,79M	+4,6%
6. Central & Northern Ukraine Depopulation Belt	12,12M	11,99M	11,39M	10,69M	10,08M	9,77M	9,66M	9,03M	-25,5%
7. Kharkiv, Dnipropetrovsk & Zaporizhzhia Industrial East	9,1M	9,02M	8,58M	8,12M	7,78M	7,62M	7,65M	7,25M	-20,3%
8. Ukrainian Black Sea Littoral	4,7M	4,68M	4,48M	4,32M	4,22M	4,18M	4,16M	4,05M	-13,8%
9. Government-Controlled Donbas	3,62M	3,58M	3,35M	3,12M	2,95M	2,86M	2,45M	2,21M	-39%
10. Occupied Donbas / ORDLO	4,6M	4,54M	4,23M	3,93M	3,7M	3,59M	2,7M	2,47M	-46,3%
11. Crimea & Sevastopol	2,53M	2,58M	2,46M	2,38M	2,34M	2,34M	2,36M	2,44M	-3,6%
12. Moscow City & Moscow Oblast Post-Soviet Hyper-Centralisation Core	15,8M	15,95M	16,65M	17,55M	18,65M	19,45M	19,95M	21,15M	+33,9%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
13. Saint Petersburg, Leningrad Oblast & Kaliningrad Exclave	7,62M	7,55M	7,42M	7,38M	7,52M	7,75M	7,95M	8,42M	+10,5%
14. Depopulating Northwest: Pskov, Novgorod, Vologda & Republic of Karelia	3,72M	3,66M	3,48M	3,26M	3,08M	3,01M	2,96M	2,8M	-24,7%
15. European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi & Nenets AO	4,32M	3,99M	3,63M	3,38M	3,15M	3,03M	2,97M	2,76M	-36,1%
16. Central Non-Black-Earth Mortality & Shrinkage Ring	12,35M	12,22M	11,75M	11,15M	10,65M	10,45M	10,35M	9,75M	-21,1%
17. Central Black-Earth Belt	7,75M	7,98M	7,85M	7,58M	7,38M	7,32M	7,3M	7,02M	-9,4%
18. Republics of Tatarstan & Bashkor-tostan	7,58M	7,85M	7,89M	7,84M	7,86M	7,91M	7,95M	7,95M	+4,9%
19. Middle Volga Oblasts	15,4M	15,68M	15,35M	14,75M	14,25M	14,05M	13,95M	13,31M	-13,5%
20. Lower Volga	7,12M	7,37M	7,27M	7,04M	6,86M	6,81M	6,77M	6,44M	-9,6%
21. Southern Russian Sunbelt Magnet	13,15M	13,95M	14,06M	13,98M	14,08M	14,24M	14,39M	14,71M	+11,9%
22. Chechnya & Ingushetia	1,32M	1,18M	1,12M	1,52M	1,69M	1,79M	1,86M	2,04M	+54,5%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
23. North Ossetia	1,92M	1,98M	2,08M	2,11M	2,12M	2,11M	2,11M	2,11M	+9,9%
24. Republic of Dagestan	1,88M	2,14M	2,45M	2,65M	2,91M	3,02M	3,11M	3,33M	+77,1%
25. Ural Industrial Belt	16,32M	16,47M	16,29M	15,87M	15,61M	15,54M	15,5M	15,7M	-3,8%
26. Khanty-Mansi	1,82M	1,81M	1,88M	1,99M	2,06M	2,13M	2,16M	2,25M	+23,9%
27. West Siberian Trans-Siberian Belt	11,15M	11,22M	11,03M	10,72M	10,54M	10,55M	10,6M	10,49M	-5,9%
28. Kuzbass Coal-Metallurgical Basin	3,18M	3,1M	2,98M	2,86M	2,76M	2,74M	2,72M	2,6M	-18,2%
29. Southern Siberian Republics: Altai, Khakassia & Tuva	1,09M	1,08M	1,06M	1,04M	1,04M	1,05M	1,06M	1,08M	-0,5%
30. East Siberian Trans-Siberian & Angara-Yenisei Corridor	6,03M	5,94M	5,71M	5,46M	5,32M	5,29M	5,29M	5,17M	-14,3%
31. Norilsk, Taymyr & Evenkia Polar De-Population Zone	380k	330k	290k	260k	235k	225k	220k	220k	-42,1%
32. Republic of Buryatia	2,41M	2,38M	2,29M	2,19M	2,08M	2,06M	2,05M	2M	-17,2%
33. Sakha Republic	1,11M	1,03M	960k	950k	958k	955k	960k	985k	-11,3%
34. Extreme Northeast Arctic Exodus Zone	1,03M	770k	630k	560k	530k	515k	505k	485k	-52,9%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
35. Southern Far East & Pacific Corridor	6,06M	5,76M	5,34M	5,1M	4,88M	4,8M	4,75M	4,58M	-24,3%
Sub-Total: Republic of Belarus (Regions 1-3)	10,25M	10,2M	10M	9,65M	9,35M	9,3M	9,3M	9,15M	-10,7%
Sub-Total: Ukraine [1991AD Borders] (Regions 4-11)	51,5M	51,23M	49,02M	46,82M	45,21M	44,52M	43,2M	41,56M	-19,3%
Sub-Total: Russian Federation [1991AD Borders] (Regions 12-35)	150,51M	151,39M	149,46M	147,19M	146,21M	146,81M	147,44M	147,37M	-2,1%
Total Post-Soviet Space (Regions 1-35)	212,26M	212,82M	208,48M	203,66M	200,77M	200,63M	199,94M	198,08M	-6,7%

Table 1. Population trajectories across the thirty-five post-Soviet regions of Belarus, Ukraine and the Russian Federation from 1991AD to 2021AD (in thousands).

As Table 1 demonstrates, the 35-region division reveals six distinct regional patterns across 1991AD - 2021AD [14][23][24].

Across the three Belarusian demographic zones (Regions 1-3, declining overall from 10,25M to 9,15M, or -10,7%), Region 1 (Minsk City and Minsk Oblast Metropolitan Magnet) expanded by +2,4% (3,4M to 3,48M) as Minsk city (1,61M to 2,01M) absorbed youth migrants from across the republic. By contrast, Region 2 (Western Belarus: Brest and Grodno Oblasts) contracted moderately by -11,9% (2,68M to 2,36M), while Region 3 (Eastern and Southeastern Belarus: Vitebsk, Mogilev and Gomel Oblasts) suffered a -20,6% contraction (4,17M to 3,31M) driven by post-Chernobyl resettlement in Gomel and Mogilev and severe rural ageing across Vitebsk [21][22].

Across the eight Ukrainian demographic and conflict zones (Regions 4-11, falling from 51,5M to 41,56M, or -19,3%), only Region 5 (Kyiv City and Kyiv Metropolitan Oblast) recorded net growth (4,58M to 4,79M, +4,6%). Region 4 (Western Ukraine: Galicia, Volhynia, Bukovina and Zakarpattia) fell by -9,1% (10,25M to 9,32M) as higher natural fertility cushioned circular EU labour emigration; Region 8 (Ukrainian Black Sea Littoral: Odesa, Mykolaiv and Kherson) declined by -13,8% (4,7M to 4,05M); Region 7 (Kharkiv,

Dnipropetrovsk and Zaporizhzhia Industrial East) contracted by -20,3% (9,1M to 7,25M, briefly rising from 7,62M in 2013AD to 7,65M in 2015AD as the primary reception belt for Donbas IDPs); and Region 6 (Central and Northern Ukraine Depopulation Belt) plunged by -25,5% (12,12M to 9,03M) due to severe rural ageing (CDR > 19-22 per 1.000 in Chernihiv and Sumy). In the conflict zones, Region 11 (Crimea and Sevastopol, annexed in March 2014AD) stood at 2,44M in 2021AD (-3,6% vs 1991AD, rising from 2,34M in 2013AD via Russian military and civilian inflow), whereas the severed Donbas coal basin suffered steep depopulation along the 2014-2021AD Minsk Line of Contact: Region 9 (Government-Controlled Donbas: Mariupol, Kramatorsk and Sievierodonetsk) fell by -39% (3,62M in 1991AD to 2,86M in 2013AD to 2,21M in 2021AD) and Region 10 (Pre-2022 Occupied Donbas: Donetsk-Makiivka-Horlivka and Luhansk-Alchevsk) collapsed by -46,3% (4,6M in 1991AD to 3,59M in 2013AD to 2,47M in 2021AD) [16][17][25].

Five non-Caucasus regions in the Russian Federation acted as metropolitan, sunbelt, hydrocarbon and petrochemical growth magnets (Regions 12, 13, 18, 21 and 26). Region 12 (Moscow City and Moscow Oblast Megaregion) surged by +33,9% (15,8M to 21,15M, +5,35M), Region 26 (Khanty-Mansi/Yugra and Yamalo-Nenets Hydrocarbon Autonomous Okrugs) grew by +23,9% (1,82M to 2,25M), Region 21 (Southern Russian Sunbelt Magnet: Krasnodar, Rostov, Stavropol and Adygea) expanded by +11,9% (13,15M to 14,71M, +1,56M), Region 13 (Saint Petersburg, Leningrad Oblast and Kaliningrad Exclave) rose by +10,5% (7,62M to 8,42M) and Region 18 (Republics of Tatarstan and Bashkortostan) increased by +4,9% (7,58M to 7,95M) [24][26][27].

In the North Caucasus republics (Regions 22, 23 and 24), Region 24 (Republic of Dagestan) recorded the highest sustained expansion in the post-Soviet space at +77,1% (1,88M to 3,33M), driven by above-replacement fertility and the Makhachkala-Kaspiysk urban boom [28]. Region 22 (Chechnya and Ingushetia) plunged from 1,32M (1991AD) to 1,12M (2000AD) during the First and Second Chechen Wars and the permanent exodus of ethnic Russians from Grozny, before rebounding through Vainakh natural increase (TFR ~2,5-2,8), the Nazran IDP expansion and federal reconstruction to 2,04M in 2021AD (+54,5%) [29][30]. Between them, Region 23 (North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia) absorbed both the 1992AD Prigorodny conflict and Georgian-South Ossetian refugee inflows, growing by +9,9% (1,92M to 2,11M).

Across the polar and Arctic depopulation zones (Regions 15, 31, 33 and 34), disaggregating the Russian North reveals the steep gradient of the post-Soviet *Severnyy ottok* ("Northern Outflow"): Region 34 (Extreme Northeast Arctic Exodus Zone: Magadan -65%, Chukotka -70% and Kamchatka) lost more than half its population at -52,9% (1,03M to 485k); Region 31 (Norilsk, Taymyr and Evenkia Polar De-Population Zone) fell by -42,1% (380k to 220k); and Region 15 (European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi and Nenets AO) fell by -36,1% (4,32M to 2,76M). By contrast, Region 33 (Sakha Republic/Yakutia) declined by only -11,3% (1,11M to 985k, recovering from a 2005AD trough of 950k) because high indigenous Sakha fertility and the rapid growth of Yakutsk (187k to 355k, +89,8%) partially offset the contraction of northern Russian mining towns [31][32][33].

Along the Trans-Siberian corridors, southern Siberian republics and European interior shrinkage belts (Regions 14, 16-17, 19-20, 25, 27-30, 32 and 35), severe ageing and rural depopulation in European Russia drove Region 14 (Depopulating Northwest and Karelia) down by -24,7% (3,72M to 2,8M) and Region 16 (Central Non-Black-Earth Mortality and Shrinkage Ring) down by -21,1% (12,35M to 9,75M), alongside declines in the Middle Volga and Finno-Ugric Republics (Region 19: -13,5%), the Lower Volga and Kalmykia (Region 20: -9,6%), the Central Black-Earth (Region 17: -9,4%) and the Ural Industrial Belt (Region 25: -3,8%). Across southern Siberia and the Pacific corridor, Region 29 (Southern Siberian Republics: Altai, Khakassia and Tuva) held virtually stable at -0,5% (1,09M to 1,08M) as high indigenous Tuvan and Altai fertility (TFR ~2,5-3) offset ethnic Russian out-migration, while Region 27 (West Siberian Trans-Siberian Belt) fell by only -5,9% (11,15M to 10,49M) thanks to the boom of Tyumen (+77,6%) and Novosibirsk (+13,7%). By contrast, coal-mine closures drove Region 28 (Kuzbass/Kemerovo Oblast) down

by -18,2% (3,18M to 2,6M), while *Zapadnyy dreyf* out-migration reduced Region 30 (East Siberian Trans-Siberian Corridor: Krasnoyarsk and Irkutsk) by -14,3% (6,03M to 5,17M), Region 32 (Buryatia and Zabaykalsky Krai) by -17,2% (2,41M to 2M) and Region 35 (Southern Far East and Pacific Corridor: Primorye, Khabarovsk, Amur, Jewish AO and Sakhalin) by -24,3% (6,06M to 4,58M) [34][35][36].

Mortality Crisis and Migration (1991-2006AD)

The defining demographic shock of the immediate post-Soviet transition (1991-2006AD) was the simultaneous collapse of period fertility and surge in adult working-age mortality across Russia, Ukraine and Belarus - a scissors-like divergence of vital rates termed the “Russian Cross” (*Russkiy Krest*) and “Ukrainian Cross” [11][12][37][38]. Figure 3 plots the crude birth rate (CBR) and crude death rate (CDR) of Russia and Ukraine (1990-2021AD) alongside the net percentage population change across all 35 regions, while Table 2 summarises decadal vital rates, life expectancy at birth (e0), cumulative natural decrease and net migration balances.

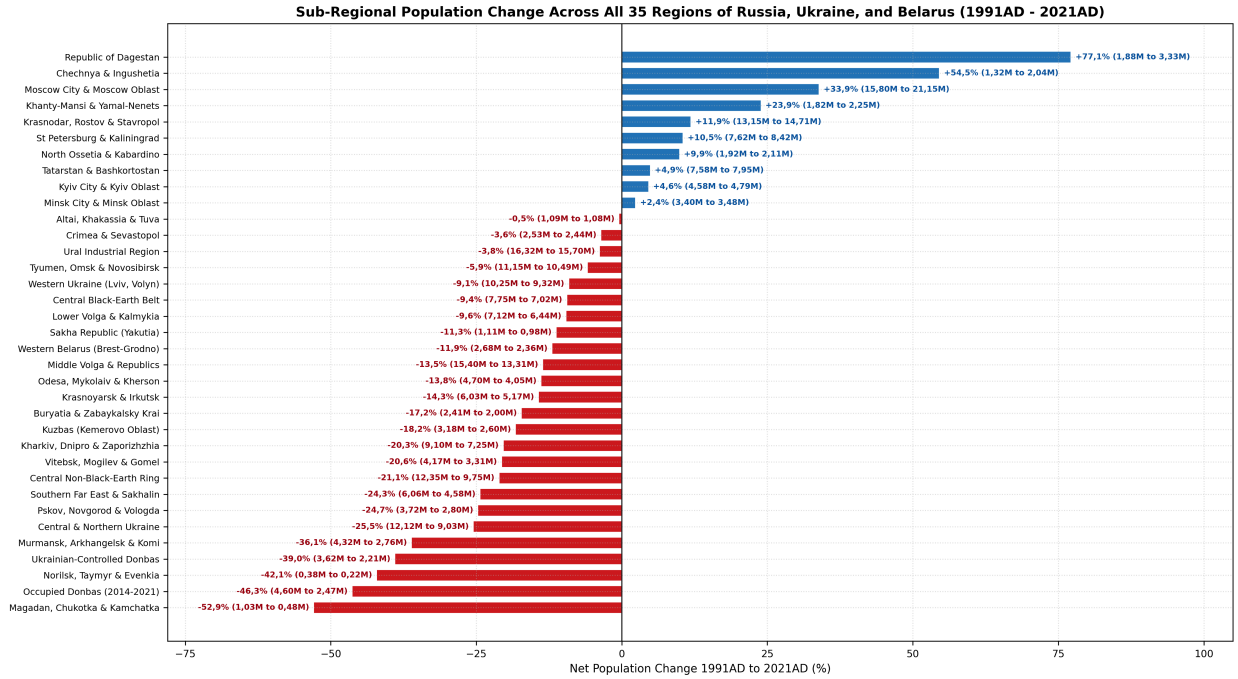


Figure 3. Crude birth and death rates per 1.000 population in Russia and Ukraine (1990AD - 2021AD) and net percentage population change across all thirty-five regions.

Country & Demographic Period	CBR/CDR (per 1.000)	Total Fertility Rate (TFR)	Male/ Female e0 (Years)	Cumulative Natural Balance	Net Migration Balance
Russian Federation: Late Soviet Baseline (1989-1991AD)	13,4/11,2	1,89	63,8/74,3	+680k	+490k
Russian Federation: Acute Transition & Hyperinflation Crisis (1992-1999AD)	9,3/14,5	1,16 (1999AD trough)	57,6/71,2 (1994AD)	-5,82M	+3,65M (CIS return wave)

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Country & Demographic Period	CBR/CDR (per 1.000)	Total Fertility Rate (TFR)	Male/ Female e0 (Years)	Cumulative Natural Balance	Net Migration Balance
Russian Federation: Second Mortality Wave & Early Recovery (2000-2006AD)	9,8/15,8	1,3	58,9/72 (2003AD)	-5,96M	+1,18M
Russian Federation: Pronatalist Recovery & Parity Window (2007-2016AD)	12,6/13,6	1,76 (2015AD peak)	66,5/77,1 (2016AD)	-1,45M (+83k in 2013-2015)	+2,65M
Russian Federation: Structural Decline & COVID-19 Shock (2017-2021AD)	10,4/13,9 (16,7 in 2021)	1,5	65,5/74,5 (2021AD)	-2,52M (-1,04M in 2021)	+980k
Ukraine [1991AD Borders]: Late Soviet Baseline (1989-1991AD)	12,6/12,1	1,85	65,6/74,9	+95k	+240k
Ukraine: Hyperinflation, Fertility Collapse & Emigration (1992-2001AD)	9,1/14,8	1,08 (2001AD world low)	61,2/72,8 (1996AD)	-2,98M	-1,15M
Ukraine: Post-2001 Depopulation & Pre-War Stabilisation (2002-2013AD)	10,1/15,6	1,46	62,3/74,1	-3,12M	-380k
Ukraine: Donbas War, Emigration & COVID-19 Shock (2014-2021AD)	9,2/15,4 (18,5 in 2021)	1,22 (1,16 in 2021AD)	65,2/74,4 (2021AD)	-2,18M (-442k in 2021)	-1,45M (Donbas/EU outflow)
Republic of Belarus: Late Soviet Baseline (1989-1991AD)	13,9/10,7	1,91	66,3/75,6	+92k	+35k
Republic of Belarus: Post-Soviet Depopulation & Rural Ageing (1992-2005AD)	9,6/13,8	1,25 (2003AD trough)	62,2/74,1 (2002AD)	-580k	+110k
Republic of Belarus: Stabilisation & COVID-19 Shock (2006-2021AD)	11,1/13,2 (16,4 in 2021)	1,45 (1,72 in 2015AD)	67,3/77,7	-480k	-95k

Table 2. Vital rates, total fertility rate, life expectancy at birth, cumulative natural balance and net migration across Russia, Ukraine and Belarus (1989AD - 2021AD).

Hyperinflation and Fertility Collapse (1991-2004AD)

Prior to the Soviet collapse, period fertility across all three East Slavic republics hovered near replacement (TFR ~1,85-2,01 in 1988-1990AD), supported by early universal marriage (mean age at first birth ~22,5 years) and state housing allocation [11][12]. Price liberalisation in January 1992AD, hyperinflation (>2.500% in Russia in 1992AD and >10.000% in Ukraine in 1993AD), the collapse of real wages by 40-60%, mass enterprise wage arrears and the closure of municipal childcare facilities triggered an immediate collapse in second and higher-order births [18][39].

In the Russian Federation (Regions 12-35), live births fell by nearly half from 2,16M (CBR = 14,6 per 1.000) in 1989AD to 1,22M (CBR = 8,3 per 1.000) in 1999AD, driving the total fertility rate down to an all-time low of 1,16 in 1999AD (0,92 in Saint Petersburg Region 13 and 1,01 in Moscow Region 12) [2][12].

In Ukraine (Regions 4-11), live births plummeted from 691k (13,4 per 1.000) in 1989AD to 376k (7,7 per 1.000) in 2001AD, with the TFR reaching 1,08 in 2001AD - then the lowest national fertility rate recorded anywhere in the world (0,89 in Kyiv Region 5 and 0,91-0,95 across the Donbas Regions 9-10 and Industrial East Region 7) [16][18].

In Belarus (Regions 1-3), live births declined from 153k (15 per 1.000) in 1989AD to 88,5k (8,9 per 1.000) in 2002-2003AD, with the TFR bottoming at 1,25 [20][21].

As Sergei Zakharov has shown, this fertility trough resulted from the intersection of an acute economic crisis with the onset of the Second Demographic Transition: young cohorts born in the 1970s and 1980s abandoned the Soviet pattern of universal childbearing at ages 20-23, shifting toward delayed marriage and older maternal schedules even as older cohorts halted parity progression after their first child [12][40].

Working-Age Male Mortality (1992-2005AD)

Simultaneously with the collapse of births, mortality surged across all three countries between 1992AD and 1994AD, followed by a brief recovery in 1995-1998AD and a second severe mortality wave after the August 1998AD rouble default that peaked in 2003AD [13][37][41]. With the repeal of the state alcohol monopoly in 1992AD, the proliferation of cheap untaxed vodka and toxic industrial alcohol surrogates (*samogon*, colognes and medicinal tinctures), psychosocial stress from rapid privatisation and the deterioration of cardiovascular care produced a sharp surge in premature deaths among men aged 25-59 [42][43][44].

Between 1991AD and 1994AD, annual deaths in Russia rose from 1,69M (11,4 per 1.000) to 2,3M (15,7 per 1.000, exceeding 21-24 per 1.000 in Region 14 Depopulating Northwest - Pskov and Novgorod - and Region 16 Central Non-Black-Earth Shrinkage Ring - Tver, Tula and Ivanovo), while Russian male life expectancy collapsed by -7,3 years in sixty months to 57,6 years in 1994AD (compared to 71,2 years for Russian women - a gender life-expectancy gap of 13,6 years) [13][37].

Following the 1998AD financial crash, Russian deaths climbed even higher to 2,37M (16,4 per 1.000) in 2003AD (male e0 = 58,9 years). Epidemiological case-control investigations, notably the Izhevsk Family Study (Region 19) led by David Leon, Vladimir Shkolnikov and Evgueni Andreev, demonstrated that hazardous binge drinking of spirits and non-beverage alcohols accounted for 43% of all deaths among working-age Russian men [38][42][43].

In Ukraine, annual deaths rose from 630k (12,1 per 1.000) in 1990AD to 792k (15,2 per 1.000) in 1995-1996AD and 765k (16,2 per 1.000) in 2005-2006AD (reaching 20-23 per 1.000 across Region 6 Central and Northern Ukraine in Chernihiv and Sumy oblasts), with male life expectancy falling to 61,2 years [38][45]. Across 1992-2006AD inclusive, cumulative excess deaths over births (natural decrease) reached -11,78M in the Russian Federation, -5,62M in Ukraine and -0,75M in Belarus - a combined vital deficit of -18,15M lives in fifteen years [2][3][20].

Repatriation and Emigration (1991-2010AD)

Why did the Russian Federation's population contract by only -2,1% across 1991-2021AD (150,51M to 147,37M) despite over -14M in cumulative natural decrease, whereas Ukraine's population plunged by -19,3% (51,5M to 41,56M)? The answer lies in three intersecting international migration streams [32][46][47].

Between 1992AD and 2002AD, the great CIS repatriation wave brought Russia a net migratory gain of +4,83M permanent settlers (peaking at +845k net in 1994AD alone, driving Russia's regional total from 150,51M in 1991AD up to 151,39M in 1995AD) from Kazakhstan (+1,85M), Central Asia (+1,42M), the South Caucasus (+820k) and the Baltics (+390k) [46][48]. These repatriates settled heavily in Region 21 (Southern Russian Sunbelt: Krasnodar, Stavropol and Rostov, rising from 13,15M to 13,95M in 1995AD), Region 17 (Central Black-Earth: Belgorod and Voronezh, 7,75M to 7,98M in 1995AD), Region 19 (Middle Volga: Samara, 15,4M to 15,68M), Region 20 (Lower Volga: Saratov and Volgograd, 7,12M to 7,37M) and Region 27 (West Siberian Trans-Siberian Belt: Novosibirsk, Omsk and Altai Krai, 11,15M to 11,22M), offsetting ~55% of Russia's natural population decline during the 1990s.

Simultaneously, the lifting of Soviet exit restrictions between 1989AD and 2005AD triggered the permanent emigration of >1,15M post-Soviet Jews (*Aliyah* to Israel alongside emigration to the United States and Germany) and >1,85M ethnic Germans (*Spätaussiedler* to the Federal Republic of Germany) [32][46]. Across Ukraine (Regions 5-8), the Jewish population fell from 486k (1989AD) to 103k (2001AD Census), particularly in Kyiv (Reg. 5), Odesa (Reg. 8), Kharkiv (Reg. 7) and Dnipro (Reg. 7); in Belarus (Regions 1 and 3), from 112k (1989AD) to 28k (1999AD); and in Russia, from 551k (1989AD) to 230k (2002AD) and 83k (2021AD), including the emigration of over half the Jewish population of Birobidzhan/Jewish Autonomous Oblast (Region 35) [1][8][15]. Concurrently, over 450k ethnic Russian Germans emigrated directly from the German national districts of Omsk Oblast (Azovo) and Altai Krai (Halbstadt) in Region 27, Saratov and Volgograd (Region 20) and the Urals (Region 25) to Germany, while another >900k Kazakhstan Germans transited through West Siberia and the Volga [46].

Ukraine initially experienced a brief positive net migration inflow of +540k in 1991-1993AD - including ~250k Crimean Tatars returning from Uzbek deportation to Region 11 (Crimea, rising from 2,53M to 2,58M in 1995AD) [15][16]. From 1994AD onward, however, deeper economic contraction turned Ukraine into a major net emigration country, losing >1,4M residents to net emigration in 1994-2001AD and 2M-3M labour migrants across Region 4 (Western Ukraine EU Migration Corridor) and Region 6 (Central Ukraine) during the 2000s and 2010s [18][49][50].

Northern Exodus and Urban Polarisation (1991-2021AD)

Alongside aggregate natural decrease, the thirty years between 1991AD and 2021AD witnessed a profound spatial polarisation of settlement across the 35 regions of Russia, Ukraine and Belarus [14][24][26][51]. Soviet central planning had dispersed tens of millions of workers into high-latitude Arctic mining towns, Siberian timber and defence settlements and single-enterprise industrial towns (*monogoroda*) through northern wage coefficients (*severnye nadbavki*) and subsidised supply lines (*severnyy zavoz*) [31][34]. When price liberalisation in 1992AD exposed the true transport and heating costs of high-latitude and single-factory settlements, two complementary migratory processes reshaped the Eurasian map: a sustained East-to-West and North-to-South internal exodus (*Zapadnyy dreyf*, or "Western Drift", and *Severnyy ottok*, or "Northern Outflow") out of Region 34 (Extreme Northeast Arctic Exodus Zone), Region 31 (Norilsk, Taymyr and Evenkia), Region 15 (European Arctic Exodus Zone) and Region 35 (Southern Far East and Pacific Corridor), and centripetal metropolitan concentration into Region 12 (Moscow Megaregion), Region 13 (Saint Petersburg and Kaliningrad), Region 21 (Southern Russian Sunbelt), Region 26 (Yugra and Yamal Hydrocarbon Arctic), Region 1 (Minsk Magnet) and Region 5 (Kyiv Metropolis)

[33][52][53]. Figure 4 contrasts expanding metropolitan and sunbelt poles against contracting Arctic, coal-mining and conflict-affected cities (1989-2021AD), while Table 3 lists representative regional and urban shifts across the 35-region division.

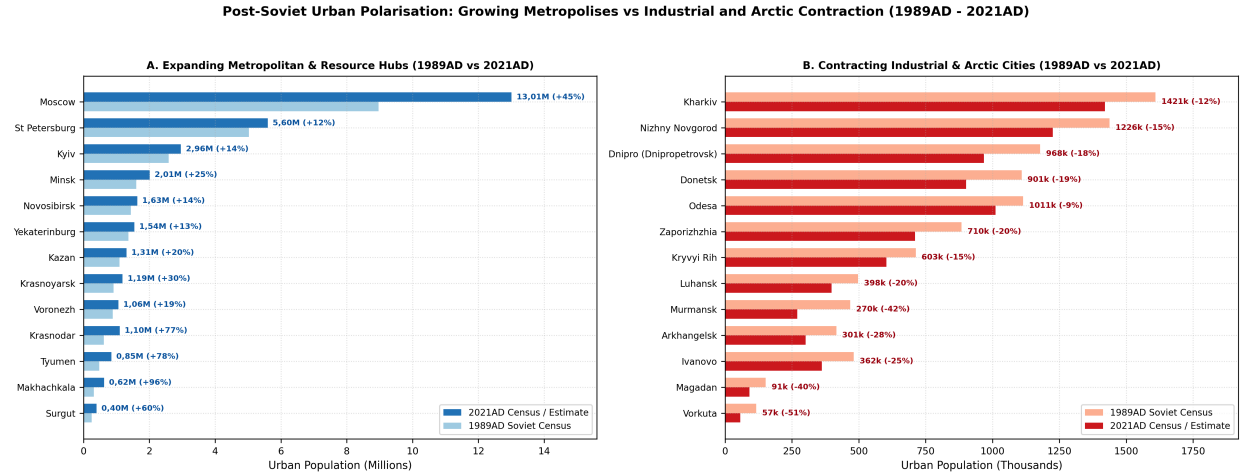


Figure 4. Divergent urban population trajectories across expanding metropolises, sunbelt capitals and contracting Arctic and industrial cities (1989AD - 2021AD).

Trajectory Category & Settlement/ Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
Extreme Northeast Exodus: Magadan, Chukotka & Kamchatka (Region 34)	1,03M (Chukotka 164k, Magadan 392k)	630k (Chukotka 54k, Magadan 183k)	530k (Chukotka 51k, Magadan 157k)	485k (Chukotka 50k, Magadan 136k)	-52,9% (Kadykchan abandoned)
Polar De-Population: Norilsk, Taymyr & Evenkia (Region 31)	380k (Norilsk aggl. 265k)	290k (Norilsk 215k)	235k (Norilsk 175k)	220k (Norilsk 175k, Igarka 3,6k)	-42,1% (Igarka -80,7%)
European Arctic Exodus: Murmansk, Arkhangelsk, Komi & Nenets AO (Region 15)	4,32M (Murmansk 468k, Vorkuta 116k)	3,63M (Murmansk 336k, Vorkuta 85k)	3,15M (Murmansk 307k, Vorkuta 71k)	2,76M (Murmansk 270k, Vorkuta 57k)	-36,1% (Vorkuta -50,9%)
Southern Far East Corridor: Primorye, Khabarovsk, Amur & Sakhalin (Region 35)	6,06M (Vladivostok 634k)	5,34M (Vladivostok 595k)	4,88M (Vladivostok 592k)	4,58M (Vladivostok 601k)	-24,3% (Amur -27,5%)

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Trajectory Category & Settlement/ Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
Yakutia Indigenous Urbanisation vs Mining Shrinkage (Region 33)	1,11M (Yakutsk 187k, Tiksi 11,6k)	960k (Yakutsk 211k, Tiksi 5,9k)	958k (Yakutsk 270k, Tiksi 5,1k)	985k (Yakutsk 355k, Tiksi 4,2k)	-11,3% (Yakutsk +89,8%)
Kuzbass Coal Restructuring: Kemerovo, Novokuznetsk & Prokopyevsk (Region 28)	3,18M (Prokopyevsk 274k)	2,98M (Prokopyevsk 225k)	2,76M (Prokopyevsk 210k)	2,6M (Prokopyevsk 178k)	-18,2% (Prokopyevsk -35%)
Ural Industrial Belt Monogoroda: Nizhny Tagil, Zlatoust & Berezniki (Region 25)	16,32M (N. Tagil 440k, Berezniki 201k)	16,29M (N. Tagil 390k, Berezniki 173k)	15,61M (N. Tagil 362k, Berezniki 156k)	15,7M (N. Tagil 339k, Berezniki 138k)	-3,8% (Yekaterinburg +13%)
Central Non-Black-Earth Shrinkage Ring: Ivanovo, Tula & Tver (Region 16)	12,35M (Ivanovo 481k, Tula 540k)	11,75M (Ivanovo 432k, Tula 481k)	10,65M (Ivanovo 408k, Tula 501k)	9,75M (Ivanovo 362k, Tula 473k)	-21,1% (Ivanovo -24,7%)
Metropolitan Growth: Moscow City & Oblast Megaregion (Region 12)	15,8M (Moscow city 8,97M)	17M (Moscow city 10,38M)	18,65M (Moscow city 11,5M)	21,15M (Moscow city 13,01M)	+33,9% (Moscow +45,1%)
Moscow Ring Boomtowns: Balashikha, Khimki, Mytishchi & Krasnogorsk (Region 12)	539k combined	586k combined	765k combined	1,22M (Balashikha 521k)	+126,5% (Balashikha +285,9%)
Baltic & Northwest Core: Saint Petersburg, Leningrad & Kaliningrad (Region 13)	7,62M (StP 5,02M)	7,42M (StP 4,66M)	7,52M (StP 4,88M)	8,42M (StP 5,6M; Murino 89k)	+10,5% (StP +11,5%)
Southern Sunbelt Magnet: Krasnodar, Sochi, Rostov & Stavropol (Region 21)	13,15M (Krasnodar 621k)	14,06M (Krasnodar 646k)	14,08M (Krasnodar 745k)	14,71M (Krasnodar 1,1M)	+11,9% (Krasnodar +77%)

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Trajectory Category & Settlement/ Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
Arctic Hydrocarbon Magnet: Surgut, Nizhnevartovsk & Novy Urengoy (Region 26)	1,82M (Surgut 248k)	1,88M (Surgut 285k)	2,06M (Surgut 307k)	2,25M (Surgut 396k, Kh.-Mansiysk 107k)	+23,9% (Surgut +59,7%)
West Siberian Trans-Siberian Boom Poles: Tyumen & Novosibirsk (Region 27)	11,15M (Tyumen 477k, Novosib. 1,44M)	11,03M (Tyumen 511k, Novosib. 1,43M)	10,54M (Tyumen 582k, Novosib. 1,47M)	10,49M (Tyumen 847k, Novosib. 1,63M)	-5,9% (Tyumen +77,6%)
Caucasus & South Siberian High-TFR Capitals: Makhachkala (Reg. 24) & Kyzyl (Reg. 29)	1,88M Reg. 24 (Makhachkala 318k; Kyzyl 87k)	2,45M Reg. 24 (Makhachkala 462k; Kyzyl 104k)	2,91M Reg. 24 (Makhachkala 572k; Kyzyl 110k)	3,33M Reg. 24 (Makhachkala 623k; Kyzyl 125k)	+77,1% Reg. 24 (Makhachkala +95,9%)
East Slavic Capitals: Kyiv Metropolis (Region 5) & Minsk Magnet (Region 1)	7,98M (4,58M + 3,4M)	7,84M (4,48M + 3,36M)	7,86M (4,52M + 3,34M)	8,27M (4,79M + 3,48M)	+3,6% (Minsk city +25,1%)

Table 3. Spatial polarisation across the thirty-five post-Soviet regions comparing northern and industrial shrinkage with metropolitan and sunbelt growth (1989AD - 2021AD).

The Northern Outflow and Siberian Corridors

Disaggregating Asian Russia and the European North into eleven regions (Region 15 and Regions 26-35) reveals why treating Siberia and the Far East as a single block obscures three distinct demographic patterns [31][32][33][34].

Across the extreme northeast and polar depopulation zones (Regions 34, 31 and 15), Region 34 (Extreme Northeast Arctic Exodus Zone: Magadan, Chukotka and Kamchatka) saw its population fall by -52,9% from 1,03M (1991AD) to 485k (2021AD): Chukotka Autonomous Okrug lost -69,5% (164k to 50k), Magadan Oblast lost -65,3% (392k to 136k) and Kamchatka Krai fell from 472k to 292k (-38,1%; Petropavlovsk-Kamchatsky 269k to 165k). Dozens of Kolyma and Chukotka mining towns - including Iultin, Valkumey and the coal town of Kadykchan (10,3k in 1989AD, abandoned after its 1996AD heating-plant explosion) - were shuttered, while Susuman shrank from 17k to 4,3k and Magadan city from 152k to 90k (-40,8%) [31][34]. In Region 31 (Norilsk, Taymyr and Evenkia Polar De-Population Zone, declining from 380k to 220k, or -42,1%), the Arctic nickel city of Norilsk contracted from 265k to 175k (-34%), Dudinka from 32k to 19,6k, timber port Igarka by -80,7% (18,8k to 3,6k) and Tura in Evenkia from 7,5k to 4,4k. Similarly, Region 15 (European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi and Nenets AO) lost -1,56M residents

(-36,1%, 4,32M to 2,76M), led by Vorkuta (116k to 57k, -50,9%), Inta (60k to 20k, -66,7%), Murmansk (468k to 270k, -42,3%), Apatity (88k to 50k), Arkhangelsk (416k to 301k) and Severodvinsk (249k to 157k) [32][35].

In the indigenous urbanisation and high-fertility southern Siberian republics (Regions 33, 29 and 32), Region 33 (Sakha Republic/Yakutia, shifting from 1,11M in 1991AD to 950k in 2005AD and 985k in 2021AD, or -11,3%) saw the exodus of >250k ethnic Russians from northern gold, tin and coal settlements (Tiksi 11,6k to 4,2k, Neryungri 82k to 57k, Ust-Nera) largely counterbalanced by high rural Sakha (Yakut) fertility and rapid migration into the capital Yakutsk, which nearly doubled from 187k (1989AD) to 355k (2021AD, +89,8%) [14][33]. Even more strikingly, Region 29 (Southern Siberian Republics: Altai, Khakassia and Tuva) held virtually constant (1,09M to 1,08M, -0,5%): in the Tuva Republic, where ethnic Russian out-migration during the early 1990s reduced the Russian share from 32% (1989AD) to 10,1% (2021AD), Russia's highest total fertility rate (TFR ~2,7-3) propelled Kyzyl up by +43,7% (87k to 125k) alongside Gorno-Altaysk (47k to 65k, +39,3%) and Abakan (156k to 185k) [5]. In Region 32 (Buryatia and Zabaykalsky Krai, 2,41M to 2M, -17,2%), indigenous Buryat rural-urban migration expanded Ulan-Ude from 353k to 438k (+24,1%) even as Zabaykalsky Krai (Chita and Krasnokamensk) contracted sharply.

Across the Arctic hydrocarbon zone and the West-East Trans-Siberian corridors (Regions 26, 27, 28, 30 and 35), oil and gas extraction in Region 26 (Khanty-Mansi/Yugra and Yamalo-Nenets Hydrocarbon Autonomous Okrugs) drove population up by +23,9% (1,82M to 2,25M), led by Surgut (248k to 396k, +59,7%), Nizhnevartovsk (245k to 283k), Novy Urengoy (95k to 107k), Noyabrsk (100k), Khanty-Mansiysk (34k to 107k, +214,7%) and Salekhard (32k to 48k). Along the West Siberian Trans-Siberian Belt (Region 27: 11,15M to 10,49M, -5,9%), rural shrinkage in Omsk and Altai Krai (Rubtsovsk and Biysk) was offset by the boom of Tyumen (477k to 847k, +77,6%), Novosibirsk (1,44M to 1,63M) and university capital Tomsk (506k to 556k). By contrast, post-Soviet mine closures in Region 28 (Kuzbass Coal-Metallurgical Basin: Kemerovo Oblast) reduced regional population by -18,2% (3,18M to 2,6M; Prokopyevsk 274k to 178k, Kiselevsk 128k to 83k, Novokuznetsk 600k to 537k) [35]. Further east, Region 30 (East Siberian Trans-Siberian Corridor: Krasnoyarsk and Irkutsk) fell by -14,3% (6,03M to 5,17M) despite Krasnoyarsk city crossing the million threshold (913k to 1,19M, +30,1%), while Region 35 (Southern Far East and Pacific Corridor: Primorye, Khabarovsk, Amur, Jewish AO and Sakhalin) declined by -24,3% (6,06M to 4,58M), with Komsomolsk-on-Amur shrinking from 318k to 238k (-25,2%) while Khabarovsk (601k to 617k) and Vladivostok (634k to 601k) absorbed regional rural youth [33][36].

Industrial Town Shrinkage and Metropolitan Growth

In European Russia, Ukraine and Belarus, Natalya Zubarevich's "Four Russias" framework captures the stark divergence between shrinking industrial and rural peripheries and booming metropolitan cores [24][26][35][54].

Across the interior shrinkage belts (Regions 14, 16-17, 19-20 and 25) - including Region 14 (Depopulating Northwest and Karelia, -24,7%), Region 16 (Central Non-Black-Earth Mortality and Shrinkage Ring, -21,1%) and Region 19 (Middle Volga and Finno-Ugric/Chuvash Republics, -13,5%) - industrial cities contracted steadily (Ivanovo 481k to 362k, Tula 540k to 473k, Rybinsk 251k to 177k, Dzerzhinsk 285k to 219k, Nizhny Novgorod 1,44M to 1,23M), while rural "spatial compression" (*szhatie prostranstva*) left over 19.400 Russian rural villages uninhabited (*derevni bez naseleniya*) by 2021AD [5][55]. In Region 25 (Ural Industrial Belt: 16,32M to 15,7M, -3,8%), the growth of Yekaterinburg (1,37M to 1,54M) and Chelyabinsk (1,15M to 1,19M) contrasted with steep declines in metallurgical and potash *monogoroda* (Nizhny Tagil -23%, Zlatoust -22,1%, Berezniki -31,3%, Orsk -30,8%).

Conversely, metropolitan concentration reshaped the major urban regions (Regions 12, 13, 21, 18, 1 and 5). Region 12 (Moscow City and Moscow Oblast Megaregion) surged by +5,35M (+33,9%, 15,8M to 21,15M), concentrating 14,4% of the Russian Federation's entire population within Moscow city (8,97M to 13,01M)

and its MKAD commuter belt (Balashikha 135k to 521k [+285,9%], Podolsk 315k, Khimki 257k, Mytishchi 255k, Korolev 228k, Lyubertsy 224k, Krasnogorsk 188k, Odintsovo 181k and Domodedovo 152k) [27][52]. Region 13 (Saint Petersburg, Leningrad and Kaliningrad) grew by +10,5% (7,62M to 8,42M; Saint Petersburg 5,6M, suburb Murino 5k to 89k, Kaliningrad 490k), Region 21 (Southern Russian Sunbelt) by +11,9% (13,15M to 14,71M; Krasnodar 621k to 1,1M [+77%], Sochi 466k, Stavropol 547k, Rostov-on-Don 1,14M), Region 18 (Tatarstan and Bashkortostan) by +4,9% (7,58M to 7,95M; Kazan 1,31M, Ufa 1,15M), Region 1 (Minsk Magnet) by +2,4% (3,4M to 3,48M; Minsk 1,61M to 2,01M) and Region 5 (Kyiv Metropolis) by +4,6% (4,58M to 4,79M; Kyiv 2,59M to 2,96M, Brovary 109k, Irpin 62k, Bucha 37k).

The Chechen Wars and North Caucasus Demography

In sharp contrast to the ageing East Slavic interior, the three disaggregated North Caucasus regions - Region 22 (Chechnya and Ingushetia), Region 23 (North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia) and Region 24 (Republic of Dagestan) - experienced both the most destructive military conflicts on Russian territory since 1945AD and the fastest long-term demographic growth across the entire 35-region division (+54,5% in Region 22, +9,9% in Region 23 and +77,1% in Region 24) [28][29][30]. Table 4 summarises the principal post-Soviet armed conflicts, casualties and forced migration flows across both the North Caucasus (Regions 22-24, 1992-2009AD) and Ukraine (Regions 9-11, 2014-2021AD).

Conflict/Territorial Event & Region	Primary Cities & Settlement Impact	Estimated War Deaths	Forced Displacement & Regional Population Shift
East Prigorodny Conflict (Region 23 North Ossetia/Region 22 Ingushetia, Oct-Nov 1992AD)	Vladikavkaz suburbs (Kambileevskoye, Chermen, Kurtat), Beslan; Nazran & Sunzha IDP camps	600 killed (plus 334 killed in Sep 2004 Beslan school siege)	40k-60k ethnic Ingush displaced from Prigorodny District to Ingushetia; 35k South Ossetian refugees into Reg. 23
First Chechen War (Dec 1994AD - Aug 1996AD) & Interwar Ichkeria (Region 22)	Grozny (400k in 1989 to <90k in Feb 1995 siege); Gudermes, Argun, Shali, Samashki	35.000 - 50.000+ military & civilian deaths	Region 22 fell 1,32M (1991) to 1,18M (1995); permanent exodus of >220k ethnic Russians from Chechnya
Second Chechen War & Counter-Insurgency (Aug 1999AD - Apr 2009AD, Region 22)	Grozny (reduced to 50k-60k in Jan 2000; rebuilt to 211k by 2002, 272k by 2010); Urus-Martan	25.000 - 40.000+ military & civilian deaths	Region 22 bottomed at 1,12M (2000) to 2,04M (2021, +54,5%); 350k Chechen IDPs fled to Ingushetia (Nazran 18k to 125k)
Dagestan Demographic Transition & Urban Boom (Region 24, 1991-2021AD)	Makhachkala (318k to 623k, +95,9%); Khasavyurt (70k to 155k); Derbent (78k to 125k); Kaspiysk (60k to 121k)	1999 Botlikh/ Kadar incursion & low-level insurgency	Region 24 surged from 1,88M (1991) to 3,33M (2021, +77,1%) via high natural increase & mountain-to-piedmont migration
Annexation of Crimea & Sevastopol (Region 11, Feb-Mar 2014AD - 2021AD)	Sevastopol (344k in 2014 to 548k in 2021 Census); Simferopol (341k); Kerch (155k); Yevpatoria (108k)	<10 direct deaths during Mar 2014 takeover	Region 11: 60k-100k Ukrainian/Tatar IDPs out vs 250k-800k Russian settler/military/resident inflow (2,34M to 2,44M)

(Continued from previous page)

Conflict/Territorial Event & Region	Primary Cities & Settlement Impact	Estimated War Deaths	Forced Displacement & Regional Population Shift
War in Donbas: Partition of Region 9 (GCA) vs Region 10 (Pre-2022 Occupied Donbas, 2014-2021AD)	427 km Contact Line: Reg. 10 Occupied (Donetsk 905k, Luhansk 399k, Makiivka, Horlivka) vs Reg. 9 Ukrainian-Held (Mariupol 431k, Kramatorsk 150k, Sievierodonetsk 101k)	14.400 killed (3.404+ civilians, 4.400 Ukrainian forces, 6.500 separatist/Russian forces)	Reg. 9 fell 2,86M (2013) to 2,21M (2021); Reg. 10 fell 3,59M (2013) to 2,47M (2021); 1,46M IDPs to Regs 5/7 + 1M to Russia

Table 4. Major armed conflicts, urban devastation, estimated war deaths and forced displacement in the North Caucasus and Ukraine from 1992AD to 2021AD.

In Region 22 (Chechnya and Ingushetia, which fell from 1,32M in 1991AD to 1,12M in 2000AD before rising to 2,04M in 2021AD, or +54,5%), the unified Checheno-Ingush ASSR at the 1989AD Soviet census contained 1,27M inhabitants (1,32M in 1991AD), including 735k ethnic Chechens, 164k Ingush and 294k ethnic Russians (23,1%), who were heavily concentrated in Grozny (399,6k) [1][29]. During the First (1994-1996AD) and Second (1999-2009AD) Chechen Wars, urban siege warfare devastated Grozny (<60k residents in early 2000AD), Gudermes, Argun, Shali and Urus-Martan, driving Region 22 down to 1,18M (1995AD) and 1,12M (2000AD) [29][30]. Over 320k-350k+ Chechen IDPs crossed into Ingushetia (joining 40k-60k ethnic Ingush earlier expelled from Prigorodny District in 1992AD), causing Nazran to swell nearly sevenfold from 18,2k (1989AD) to 125,1k in the 2002AD Census (122k in 2021AD alongside new republic capital Magas at 15k) [8][30]. Over 250k+ ethnic Russians permanently departed Chechnya and Ingushetia for Region 21 (Southern Russian Sunbelt), reducing the Russian share in Chechnya to 1,2% (2021AD). Conversely, post-2004AD federal reconstruction rebuilt Grozny (211k in 2002AD to 272k in 2010AD to 329k in 2021AD), while high Vainakh fertility (TFR ~2,5-2,8) and a young age structure (median age ~24 years) propelled Region 22 to 2,04M (1,51M in Chechnya and 510k in Ingushetia in the 2021AD Census) [5].

In Region 23 (North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia, growing from 1,92M to 2,11M, or +9,9%), the East Prigorodny armed conflict outside Vladikavkaz in October-November 1992AD killed ~600 people and displaced 40k-60k ethnic Ingush from Prigorodny District into Ingushetia (Region 22), while North Ossetia (Vladikavkaz 300k to 296k, and Beslan 33k to 36k, site of the September 2004AD school siege) simultaneously absorbed >35k ethnic Ossetian refugees fleeing the 1991-1992AD South Ossetian war in Georgia [28][30]. Across Kabardino-Balkaria (Nalchik 235k to 247k, Prokhladny 58k to 60k) and Karachay-Cherkessia (Cherkessk 113k to 113k), moderate indigenous natural increase offset gradual ethnic Russian out-migration, stabilising Region 23 at 2,11M (2005-2021AD).

In Region 24 (Republic of Dagestan, expanding from 1,88M to 3,33M, or +77,1%), the population nearly doubled across 1991-2021AD (+1,45M), driven by sustained natural increase (TFR ~1,9-2,4 coupled with low adult mortality, CDR ~5,5-7 per 1.000) and large-scale internal resettlement of Avar, Dargin, Lezgin, Kumyk and Lak mountain villagers down to the Caspian coastal plain [5][28]. Consequently, Makhachkala nearly doubled from 318k (1989AD) to 623k (2021AD, +95,9%, or >1M in its wider agglomeration), while Khasavyurt surged by +121,4% (70k to 155k), Kaspiysk by +101,7% (60k to 121k) and Derbent by +60,3% (78k to 125k).

Crimea, the Donbas War and COVID-19 (2014-2021AD)

The final seven years of the 1991-2021AD period were shaped by the armed conflict and territorial partition of Crimea (Region 11) and the Donbas (Regions 9 and 10) in 2014-2021AD, followed by the severe 2020-2021AD COVID-19 excess mortality wave [17][25][56][57].

Annexation of Crimea and Sevastopol

Following the Russian annexation of the Autonomous Republic of Crimea and Sevastopol (Region 11) in February-March 2014AD, the peninsula's benchmark population shifted from 2,34M (2013AD) to 2,36M (2015AD) and 2,44M (2021AD benchmark; 2,48M in the October 2021AD Russian Census) [5][10]. Between 60k and 100k Ukrainian citizens and Crimean Tatars relocated as IDPs to mainland Ukraine (Regions 4, 5 and 8), whereas 250k-350k+ permanent Russian military personnel, federal civil servants and mainland retirees (and up to 500k-800k cumulative gross arrivals including construction workers and unregistered temporary residents across 2014-2021AD) moved onto the peninsula, driving Sevastopol up from 344k (2014AD) to 548k (2021AD) alongside Simferopol (341k), Kerch (155k) and Yevpatoria (108k) [17][50].

The War in Donbas

In April 2014AD, armed conflict erupted across Donetsk and Luhansk oblasts (6,45M combined residents in 2013AD), severing the integrated Donbas coal-and-steel conurbation along a fortified 427 km Line of Contact (Minsk II) into two distinct demographic zones [3][17][25].

Region 10 (Pre-2022 Occupied Donbas, which fell from 4,6M in 1991AD to 3,59M in 2013AD and 2,47M in 2021AD, or -46,3%) encompassed the occupied urban core of Donetsk (965k in 2013AD to ~905k nominal or <850k de facto in 2021AD), Luhansk (426k to ~399k), Makiivka (353k to ~335k), Horlivka (256k to ~235k), Alchevsk (111k to 105k), Yenakiieve (82k to 75k) and Kadiivka/Stakhanov (77k to 72k). Between 2013AD and 2015AD alone, Region 10 lost -890k inhabitants (-24,8%) through mass flight, followed by acute natural decrease (CDR > 21 per 1.000 vs CBR < 5,5 per 1.000) that reduced its present population to 2,47M by 2021AD [17].

Region 9 (Government-Controlled Donbas, declining from 3,62M in 1991AD to 2,86M in 2013AD and 2,21M in 2021AD, or -39%) encompassed Ukrainian-controlled Mariupol (459k to 431k), provisional Donetsk oblast seat Kramatorsk (163k to 150k), Sloviansk (117k to 106k), provisional Luhansk oblast seat Sievierodonetsk (109k to 101k), Lysychansk (104k to 93k), Kostiantynivka (76k to 68k), Bakhmut (77k to 71k) and Pokrovsk (65k to 60k).

Up to 31 December 2021AD, the Donbas conflict killed ~14.400 people and displaced over 2,5M residents. Roughly 1,46M registered IDPs relocated within government-controlled Ukraine - above all into neighbouring Region 7 (Kharkiv, Dnipropetrovsk and Zaporizhzhia, whose population briefly rose from 7,62M in 2013AD to 7,65M in 2015AD) and Region 5 (Kyiv Metropolis, rising from 4,58M to 4,66M in 2015AD) - while ~1M refugees and migrants moved into the Russian Federation (Region 21 Rostov/Krasnodar, Region 17 Voronezh/Belgorod and Region 12 Moscow) and Belarus (Regions 1-3) [17][18][25].

Excess Mortality from COVID-19 (2020-2021AD)

After a brief 2013-2015AD vital-rate parity window in Russia (+83k natural increase), structural birth decline returned after 2016AD as the small cohorts born in the 1990s entered childbearing age, followed in 2020-2021AD by the COVID-19 pandemic [40][56][57][58].

Across the Russian Federation (Regions 12-35), registered deaths surged from 1,8M (2019AD) to 2,14M (2020AD) and 2,44M (2021AD, 16,7 per 1.000), producing 1,07M - 1,15M cumulative excess deaths, driving

annual natural decline to a post-WWII record of -1,04M (-1,04M) in 2021AD alone and reducing Russia's total population across Regions 12-35 from 148,07M (2018AD) to 147,37M (2021AD) [57][58].

In Ukraine (Regions 4-11), ~185k COVID-19 excess deaths pushed annual natural decline to -442k in 2021AD (CDR = 18,5 per 1.000 vs CBR = 7,3 per 1.000), reducing Regions 4-11 from 42,33M (2018AD) to 41,56M (2021AD) [3][18]. In Belarus (Regions 1-3), ~70k-85k excess deaths and post-2020AD emigration reduced the population from 9,25M (2018AD) to 9,15M (2021AD), bringing the entire 35-region post-Soviet space to 198,08M (198,08M) on the eve of the February 2022AD invasion [4][56].

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